University of Dhaka

Affiliated Faridpur Engineering College

Department of Computer Science and Engineering

1st year semester 1st, 1st In course Examination -2024 (Session: 2023-24)

Time: 50 mints **Marks: 20**

1. (a) Define the limit of a function using $(\delta-\varepsilon)$ .Using $(\delta-\varepsilon)$ definition prove that $\lim_{x\to 3}\frac{x^2-9}{x-3}=6$ 1+2

(b) Evaluate (any two) (*i*) $\lim_{x\to 0}(1+x)^{\frac{1}{x}}$ (*ii*) $\lim_{x\to 0}\left(\frac{\tan x}{x}\right)^{\frac{1}{x}}$ (*iii*) $\lim_{x\to \pi/2}\frac{e^{\tan x}-1}{e^{\tan x}+1}$ 1+1

(c) If $\varphi(x)=\begin{cases} x^2, when & x<1 \\ 2.5, when & x=1 \\ x^2+2, when & x>1 \end{cases}$ Dose $\lim_{x\to 0}\varphi(x)$ exist? 2

(d) If $x^y+y^x=a^b$ then determined $\frac{dy}{dx}$. 3

2. (a) If the function $f(x)$ is differentiable at $x=a$ then the function is continuous at that point. 2

(b) If $f(x)=\begin{cases} 1, & when \quad x<0 \\ 1+\sin x, & when \quad 0\le x<\frac{\pi}{2} \\ 2+\left(x-\frac{\pi}{2}\right)^2, & when \quad x\ge 0 \end{cases}$, then show that $f(x)$ is continuous at $x=\frac{\pi}{2}$. 5

but not differentiable at $x=0$.

(c) Differentiate $\tan^{-1}\frac{x}{\sqrt{1-x^2}}$ with respect to $\sec^{-1}\frac{1}{2x^2-1}$ 3